

Muhammad Ramis

Offensive Security Researcher

AI Safety | Red Teaming | Cloud Infrastructure Security

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PROFILE

Offensive Security Researcher with 8+ years of hybrid experience, cloud-native architectures, and advanced vulnerability hunting. Deep expertise in exploiting complex systems, including prompt injection, model manipulation, and dynamic UI component abuse in agentic workflows. Proven track record of designing sophisticated red and purple team exercises, building automated offensive tooling, and collaborating seamlessly with detection engineering teams to harden critical infrastructure.

CORE SKILLS

AI / ML Security: Prompt Injection, Data Leakage, Model Manipulation, Dynamic UI Abuse, Agentic Workflow Pentesting, AI Safety, Threat Modeling, LLM Vulnerability Hunting

Offensive Tooling: Exploit Development, Automated Offensive Workflows, Red / Purple Teaming, Custom Malware Analysis, Penetration Testing, GitHub Security Best Practices

Infrastructure & Cloud: Custom Kubernetes, Cloud-native Architectures (AWS, GCP, Azure), CI/CD Pipelines, Containerized Environments, macOS / Linux Internals

Development & Data: Python, React, Go, Rust, Data Science Toolchains, AI/ML Infrastructure, PyTorch, SRE Practices, Detection Engineering Collaboration

EXPERIENCE

Lead Engineer | Sorvox Groups | 2022 - 2024

Stack: Python, Custom Kubernetes, AWS, PyTorch, LLM Frameworks

- Proactively hunted vulnerabilities in agentic multi-LLM workflows, discovering and exploiting 45+ critical vulnerabilities including advanced prompt injection and model data leakage.
- Designed and executed complex red team exercises targeting cloud infrastructure and foundational models, achieving a 98% success rate in simulating highly sophisticated adversarial scenarios.
- Built and automated custom offensive Python tooling to stress-test data science toolchains, scaling impact and reducing manual testing overhead by 75%.
- Partnered with defensive teams to translate offensive insights into actionable detection engineering metrics, decreasing mean-time-to-remediate (MTTR) for AI-specific flaws by 60%.

Department Lead IT | Apricart | 2020 - 2022

Stack: GCP, Kubernetes, CI/CD, React, Go, GitHub Security

- Conducted in-depth penetration testing across cloud-native architectures, React-based dynamic UI interfaces, and complex microservices, identifying 30+ critical trust boundary violations.
- Architected purple team workflows simulating modern technology stack breaches, increasing organizational detection capabilities across custom Kubernetes environments by 85%.
- Automated CI/CD and GitHub security pipelines using Go and Python, preventing 100% of high-risk supply chain vulnerabilities and hardcoded secrets from reaching production.
- Communicated high-impact risks to technical and non-technical stakeholders, effectively driving the prioritization of critical mitigations for $2M +$ *in infrastructure assets*.

Full Stack Developer | Creatie Drop DMCC | 2018 - 2019

Stack: AWS, Linux Internals, Bash, Python, Data Science Infrastructure

- Exploited subtle vulnerabilities in complex custom deployments and data science environments, securing highly sensitive user data and proprietary models against extraction.
- Contributed an attacker-informed perspective to organizational risk assessments, completely reshaping the threat modeling framework for cloud-native product deployment.
- Authored robust exploit chains for macOS and Linux internal testing, highlighting critical privilege escalation paths and misconfigurations in core networking components.
- Collaborated with Site Reliability Engineering (SRE) teams to harden infrastructure, implementing automated anomaly detection scripts that caught 15+ unauthorized access attempts.

Startup Venture | Lead Engineer | Security | Chatcloud | 2016 - 2018

Stack: Python, React, AWS, Linux, OWASP, Secure SDLC

- Applied a builder's mindset to integrate security directly into product development, writing robust Python and React components while acting as the primary offensive tester.
- Built custom automated scanners for web application vulnerabilities, reducing successful pre-production exploitation vectors by 90% across all dynamic user-facing interfaces.
- Bridged the gap between software engineering and security architecture, ensuring secure-by-default configurations for early-stage cloud deployments.
- Championed security awareness, distilling technical nuances into actionable security guidelines and directly mentoring 10+ software engineers on threat mitigation.

CERTIFICATIONS

OSCP - Offensive Security Certified Professional | 2025

Practical penetration testing certification covering exploitation, privilege escalation, reporting, and remediation-focused security assessment.

OSED - Offensive Security Exploit Development | 2026

windows user-mode exploit development exam in june.

EDUCATION

MSc Cyber Security and Artificial Intelligence | The University of Sheffield, UK | Sep 2024 - Nov 2025

Distinction; dissertation on post-quantum blind signatures with Python, Rust, Node.js, gRPC, and REST tooling.

BSc Computer Science | PAF KIET, Karachi, Pakistan | Aug 2014 - Dec 2018

First Class; final project: React Native carpooling platform with web admin, GPS, chat, notifications, and Laravel backend.

ACHIEVEMENTS

- **OffSec The Gauntlet Echo Response S1:** Placed 4th in a forensics CTF by analysing provided evidence files and answering challenges under time pressure.
- **ShefESH Capture The Flag:** Won 1st place by exploiting web vulnerabilities and documenting remediation steps.
- **ACM-ICPC Regional Finals:** Placed top 10 and fastest solver, solving algorithmic problems against 40+ regional teams.
- **Logistic Hunt Speed Programming:** Led team to 1st place and served as COMBAT 18 technical lead for 200+ participants.
- **Apricart E-Stores:** Earned Employee of the Month; finalist in 15+ programming contests while mentoring peers.
- **COMBAT 2015 - 2017:** Co-led Speed Programming and Algomania training sessions and events for 100+ participants.